

Research Article

Music Learning Among Digital Native Learners in Thai Higher Education

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Abstract

Higher music education worldwide is experiencing a profound digital transformation, redefining how students learn, access, and engage with musical knowledge. This shift has intensified with the rise of Generation Z—often termed “digital natives.” Having grown up immersed in technology, these students demonstrate fluency in digital tools, expect immediacy, and navigate multiple information sources with ease. Their behavior differs from earlier previous generations, creating both challenges and opportunities for higher education music programs. In Thailand, music curricula remain grounded in Western classical and Thai traditional practice, typically delivered through hierarchical, instructor-centered models. However, today’s students interact with music through globalized and digitally mediated environments, revealing a gap between traditional pedagogy and contemporary learning. While previous research explored specific tools such as online tutorials or mobile applications, no comprehensive framework explains how Thai digital natives’ approach music learning in higher education.

This study aimed to construct a grounded theory explaining how digital-native music students in Thai universities learn. Its objectives were to identify causal conditions, contextual factors, strategies, and consequences shaping their learning process, providing a framework to inform curriculum design, pedagogy, and policy.

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A grounded-theory approach was used to inductively build theory from participants' experiences. Data came from five focus-group discussions with undergraduate music students born in the late 1990s–early 2000s, represented true digital natives across Thailand's five regions and four interviews with veteran instructors. Participants, Purposive and snowball sampling ensured regional and institutional diversity. Discussions explored formal and informal, technology-mediated learning, while interviews examined pedagogical perspectives. All sessions were ethically approved, recorded, transcribed, and systematically analyzed.

Following Strauss and Corbin's coding procedures, open coding generated over 150 initial codes. Axial coding organized them through causal–context–strategy–consequence linkages, and selective coding integrated categories into a coherent theoretical framework. Constant comparison and triangulation across student and instructor data enhanced validity, and expert peer review ensured credibility.

The core phenomenon was blended self-directed music learning, in which students combined formal instruction with informal digital learning. They demonstrated strong agency through YouTube, mobile apps, and digital audio tools to extend or substitute traditional methods. Strategies included slowing playback, looping passages, recording and reviewing practice sessions, and sharing materials with peers. Learning extended beyond classrooms, allowing pursuit of personal interests. Three major causal conditions supported this process: early digital exposure, personal musical interests diverging from approved programs, and technological affordances such as smartphones and internet access. The COVID-19 pandemic further normalized digital music-learning practices.

Institutional curricula and cultures shaped whether students sought resources beyond coursework. Teachers' attitudes influenced whether digital learning was validated or marginalized. Infrastructure—including internet quality, device access, and software availability—either facilitated or constrained engagement. Peer networks enhanced collaboration, creating interactive learning communities.

Blended self-directed learning produced improved technical skills, creativity, autonomy, and motivation through interest-driven engagement. Many students exceeded instructors' expectations by mastering advanced repertoire or developing innovative cross-genre projects, such as merging Thai traditional instruments with electronic production.

However, risks included shallow learning, distraction, and neglect of fundamental skills. Consequently, teachers' roles evolved from authoritative transmitters to facilitators, curators, and mentors who help students evaluate online resources, structure practice, and connect informal exploration with formal study.

The conclusion, digital-native music students in Thai higher education represent a new paradigm of proactive, technology-mediated, self-directed learning. The grounded theory positions *blended self-directed music learning* as the central process through which students integrate digital exploration with formal instruction, guided by personal interests and enabled by technology within institutional and peer contexts. Curricula should become flexible and student-centered, incorporating digital creativity, popular genres, and cross-genre innovation alongside Western and Thai classical foundations. Pedagogically, instructors should adopt blended approaches, curate reliable online resources, and explicitly teach digital literacy and critical evaluation. Institutions must enhance technological infrastructure and create assessment systems that value informal and digitally mediated outcomes.

At the policy level, national strategies for digital integration in arts education should be established, including the creation of Thai-language digital resources to enhance accessibility and equity. Collectively, these initiatives will help Thai music education remain relevant, inclusive, and innovative—preparing students for creative and connected musical futures in the digital era.

Keywords: Higher Education / Digital Natives / Music Program / Grounded Theory /
Blended Self-Directed Music Learning

Higher education worldwide is undergoing a profound digital transformation that reshapes teaching and learning practices. Contemporary university students largely belong to Generation Z, often described as “digital natives”—native speakers of the digital language of computers, video games and the Internet.¹ This generation values individual expression,

¹ Marc Prensky, “Digital Natives, Digital Immigrants, Part 1,” *On the Horizon* 9, 5 (2001): 1.

inclusivity, and open dialogue, while being more pragmatic and analytical than earlier generations.² In music education, the rise of digital natives coincides with rapid technological advancement that is redefining how young people engage with music. Digital technologies have altered how music is created, shared, and learned, giving rise to what scholars term an “unprecedented renaissance of social music making” through apps, online platforms, and collaborative networks. Consequently, higher education institutions are urged to adapt curricula and pedagogy to these emerging realities.

In Thailand, these global shifts are keenly experienced within higher education music programs that balance traditional Thai and Western classical music foundations with contemporary practices. Thai universities now face the challenge of meeting the learning styles of “digital citizen” students—individuals who have interacted with digital technology from early childhood. Digital citizens are accustomed to constant connectivity and on-demand learning opportunities,³ signifying a paradigm shift in music education. Instructors increasingly move from serving as primary knowledge providers to becoming facilitators or mentors guiding technologically adept learners. Nevertheless, disparities in digital infrastructure—such as unstable internet access or limited quality devices—pose barriers to equitable implementation. The COVID-19 pandemic further accelerated online teaching adoption, intensifying the need for robust digital competencies and infrastructure in Thai higher education.

Despite the clear influence of digital technology on contemporary music learners, there remains no formal theory explaining how digital-native students learn music holistically in higher education, particularly in Thailand. Studies on digital natives have focused mainly on general education contexts,⁴ while music education research has examined isolated tools,

² Kim Parker and Ruth Igielnik, “On the Cusp of Adulthood and Facing an Uncertain Future: What We Know about Gen Z So Far,” accessed May 14, 2020, <https://www.pewresearch.org/social-trends/2020/05/14/on-the-cusp-of-adulthood-and-facing-an-uncertain-future-what-we-know-about-gen-z-so-far/>.

³ Peter Gouzouasis and Danny Bakan, “The Future of Music Making and Music Education in a Transformative Digital World,” *UNESCO Observatory E-Journal* 2, 2 (2011): 2.

⁴ Marc Prensky, 2.

such as mobile applications for listening practice among secondary students.⁵ To the authors' knowledge, this study is the first grounded theory research in Thailand to inductively theorize digital-native music learning in higher education.

The study aims to construct a grounded theory that explains the music learning process for this group, identifying key categories (causal conditions, context, strategies, and consequences) and the relationships between them. By achieving these objectives, the contribution is to a theoretical model that can inform educators and policymakers on how to better align music education practices.

Methodology

This study employed a constructivist grounded theory methodology to investigate music learning in Thai higher education. Data collection and analysis proceeded iteratively using purposive, snowball, and theoretical sampling. The constant comparative method guided ongoing analysis, with emerging categories informing subsequent data collection and questioning. Researchers maintained theoretical sensitivity by grounding interpretations in participants' data while critically reflecting on prior assumptions. This iterative process continued until theoretical saturation was achieved, indicating that additional data no longer generated novel conceptual insights.

Participants and Data Collection

This study, grounded in the constructivist grounded theory perspective, was conducted within Thai higher education music programs. Purposive sampling was used to recruit participants for in-depth interviews, while snowball sampling identified participants for focus groups. Participants included undergraduate music students (digital natives) and music instructors from five purposively selected universities representing Thailand's major regions (Northern, Northeastern, Central, Eastern, and Southern), capturing diverse institutional and

⁵ Sunggi Cho, Yoomee Baek, and E. J. Choe, "A Strategic Approach to Music Listening with a Mobile App for High School Students," *International Journal of Music Education* 37, 1 (2019): 133–134, <https://doi.org/10.1177/0255761418819016>.

digital learning contexts in higher music education. The inclusive criteria for student participants were: (1) current enrollment in an undergraduate music program; and (2) status as third-year, fourth-year or higher-level students, reflecting digital-native characteristics. The inclusive criteria for instructor participation included full-time employment as music lecturers in higher education institutions, a minimum of five years of teaching experience, and direct responsibility for instructing undergraduate music students.

The data were collected through five semi-structured focus groups with students (4–10 per group) and four in-depth interviews with instructors. Discussions explored experiences of music learning and technology use in formal and informal contexts. Two interviews were held face-to-face and two via Zoom. Each session lasted 60–90 minutes, was recorded with consent, transcribed, and anonymized. Ethical approval and informed consent were obtained before data collection.⁶ This study received approval from the Institutional Review Board (IRB) at Mahidol University (Approval No. 2022/031.2802). Written informed consent was obtained from all participants prior to data collection.

Data were gathered through five semi-structured focus group discussions with students from universities representing Thailand’s five major regions (North, Northeast, Central, East, and South). Each group comprised 4–10 participants from any area of music program such as music performance, music education and musicology. The discussions explored how students learned music both inside and outside the classroom and how digital tools supported their learning, with many noting YouTube as their “first resort.”

Additionally, four in-depth interviews were conducted with instructors from different regions to examine their perceptions of students’ learning behaviors and the integration of technology in teaching. Two interviews were held on campus and two via Zoom due to pandemic constraints. Each session lasted 60–90 minutes, was audio-recorded with consent, transcribed, and anonymized. Ethical approval and informed consent were obtained prior to data collection.

⁶ British Educational Research Association, *Ethical Guidelines for Educational Research*, 4th ed. (London: BERA, 2018), 8-12, accessed October 15, 2025, https://www.bera.ac.uk/wp-content/uploads/2018/06/BERA-Ethical-Guidelines-for-Educational-Research_4thEdn_2018.pdf.

Data Analysis Procedure

The analysis followed the classic grounded theory coding process, with qualitative data organized using Microsoft Word software for transcript management. Figure 1 presents the overall flow of data collection and analysis. Following transcription, the process began with open coding, during which interview and focus group transcripts were broken down into discrete units of meaning. In this phase, the data were examined line by line to identify significant concepts and to assign preliminary codes (e.g., “learning via YouTube,” “difficulty with music theory app,” “teacher as coach”). Open coding enabled relevant concepts to emerge without imposing predetermined categories. More than 150 initial codes were generated, with careful attention paid to participants’ words.

The data collection and coding process in the grounded theory study. Focus group discussions with students and interviews with instructors provided raw qualitative data. Using Strauss and Corbin’s coding techniques conducted open coding to identify initial concepts, axial coding to develop categories and linkages, and selective coding to integrate the categories around a core phenomenon, yielding an emergent grounded theory of music learning for digital natives.

In the axial coding phase, the relationships among the open codes were examined and clustered into higher-level categories and subcategories. Following Strauss and Corbin’s approach, axial coding is described as “an advanced procedure for interconnecting the categories.” Techniques such as constant comparison and memo-writing were employed to relate codes to one another. For example, multiple open codes—including “prefers visual learning (tutorial videos) over text,” “short attention span in class,” and “uses smartphone during practice”—were grouped into a subcategory labeled learning style shaped by digital media. This subcategory was subsequently connected to a broader category, Learner Characteristics. Similarly, codes associated with teachers’ actions (e.g., “assigns online research,” “incorporates music apps in assignments”) were grouped under Instructor Strategies.

Strauss and Corbin's paradigm model⁷ was applied as an analytic tool during axial coding to identify causal conditions, contextual factors, intervening conditions, actions/interactions, and consequences within the data. This framework facilitated the organization of the emerging theory. For instance, when students expressed that "I learn a lot of songs by myself from the internet because my classroom lessons cover mostly classical pieces I'm not interested in," the analysis identified the causal condition (limited relevance of curriculum content), the action/strategy (self-directed online learning), the context (easy internet access, personal musical taste), and the consequence (increased motivation to learn preferred songs but reduced engagement with formal class material).

In the phase of selective coding, a core category was identified as represented the central phenomenon of the study, with all other categories systematically related to this core. Selective coding entailed the integration and refinement of the emerging theory, producing a narrative framework that interconnected the categories into an explanatory model. At this stage, all transcripts were revisited to ensure that the developing theory remained grounded in the data and to identify any disconfirming evidence.

The core category that emerged was the Digital Native Music Learning Process, encapsulating how students assumed responsibility for their music learning through technology both within and beyond formal education. Around this central category, the analysis integrated the causal conditions underlying the process, the contextual factors shaping it, the action/interaction strategies employed by students and instructors, and the resulting consequences. To enhance credibility and mitigate researcher bias, the developing theory was reviewed with a peer debriefer (an independent qualitative researcher). Triangulation of data sources—comparing the perspectives of students and instructors—further strengthened the trustworthiness of the findings.

Through these steps, a coherent grounded theory was established, presented in the following section along with its key categories and interrelationships.

⁷ Juliet Corbin and Anselm L. Strauss, *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*, 3rd ed. (Thousand Oaks, CA: SAGE Publications, 2008), 97, <https://doi.org/10.4135/9781452230153>.

Research Results

Overview of the Grounded Theory: The central phenomenon constructed from the data is a technology-mediated, self-regulated learning process by which digital native students engage with music. Digital native music students in Thai higher education actively integrate formal classroom learning with informal digital resource-based learning to fulfill their musical learning needs. This process is driven by certain antecedents.

Core Phenomenon – “Blended Self-Directed Music Learning”: At the heart of the theory is the observation that digital native students do not rely solely on traditional, instructor-led instruction; rather, they supplement and sometimes avoid formal methods with self-directed learning using digital tools. Blended Self-Directed Music Learning, identified as the core category, captures the manner in which students combine diverse modes of learning. For example, a student might learn classical pieces through formal lessons in class, while using YouTube tutorials and mobile apps to learn pop songs or to practice specific techniques on their own. One focus group participant described this blend succinctly: “In class I learn theory and the assigned repertoire, but on YouTube I learn the songs I want and new tricks from various creators.” This illustrates how the learner combines formal curriculum with informal, interest-driven exploration. The core phenomenon captures the agency of students in steering their learning and the ubiquity of digital technology as an enabler.

Causal Conditions: Several causal conditions shape this phenomenon. 1) Early Digital Exposure: As true digital natives, students have grown up with smartphones, computers, and the internet as natural learning tools. This lifelong immersion fosters expectations of immediacy and interactivity. Many enter university already self-taught in basic instruments through YouTube or Google, making them confident, independent learners. 2) Personal Interest and Individualized Goals: Students often pursue genres or instruments beyond the formal curriculum, such as jazz, K-pop, or EDM. This mismatch motivates them to seek self-directed online learning. One vocal student noted following Broadway coaches on YouTube to complement classical training. 3) Technological Affordances: Widespread internet access, smartphones, and music apps in Thailand enable continuous, flexible learning, allowing students to explore, practice, and share music beyond institutional boundaries. 4) COVID-19 pandemic: It further accelerated this trend. With in-person classes suspended, students and

instructors adapted to online platforms, using Zoom lessons, digital sheet music, and recording assignments. This period rapidly enhanced digital literacy and normalized technology use in music education, with many practices continuing even after campuses reopened.

Contextual Conditions: The context in which digital native music learning occurs includes both the educational environment and the technological infrastructure. Several key contextual factors emerged:

1. **Institutional Curriculum and Culture:** Thai higher education's curriculum structure and expectations form an important backdrop. Some institutions maintain conservative curricula focusing on classical training, which, as mentioned, can leave gaps between what is taught and students' personal musical interests. In such a context, students feel the need to look beyond the curriculum to satisfy their musical curiosity. Conversely, in programs where instructors are more open to contemporary genres or student input, students reported feeling more engaged and blending their learning more seamlessly (formal and informal learning reinforcing each other). The predominant culture – whether innovation-oriented or tradition-oriented – thus influences how much students rely on outside digital learning.

2. **Teacher Support and Attitude:** Instructors play a pivotal role in shaping technology use in the classroom. Some actively support it by recommending YouTube channels, sharing e-learning materials, or assigning online-based projects, thereby validating and guiding students' self-directed learning. Others, however, express hesitation or concern about online learning causing “bad habits.” In such cases, students continue exploring independently, often without teacher oversight. Consequently, instructor attitudes determine whether technology integration fosters a cohesive blend of formal and informal learning or leaves students' self-learning efforts more isolated and disconnected from classroom guidance.

3. **Technological Infrastructure:** The availability and quality of technology form a key contextual condition. Internet access and device quality vary across Thai universities. Students at regional campuses reported issues like slow Wi-Fi or outdated music labs, limiting digital learning opportunities. In contrast, those with better facilities or personal devices experienced greater success. Language also matters—many online tutorials and forums are in English, giving students with higher proficiency an advantage, while others rely on limited Thai resources.

4. Peer Environment: Social context further shapes learning. Students often exchange digital materials, tutorials, and advice within peer groups, reinforcing collaborative and technology-driven music learning.

Action/Interaction Strategies: Given the causal pressures and operating within the above contexts, digital native music students employ a variety of strategies to learn. The data revealed a rich repertoire of actions and interactions, which were distilled into several major strategy categories:

1. Manipulate Online Resources: Nearly all students reported using online platforms to learn music, though approaches varied. The most common strategy was watching YouTube tutorials, performances, and demonstrations, often following specific instructors or channels for skills such as guitar or vocal techniques. For learning new songs, students searched for cover versions or step-by-step tutorials. Beyond YouTube, they accessed educational websites and apps like Coursera, Udemy, or Yousician, and used Google or Wikipedia to quickly check music theory concepts. This behavior reflected a self-directed attitude toward learning, summarized by one student's remark: *"Why ask the music professor something I can find in a second online?"*

2. Self-Paced Practice and Iterative Learning: Another strategy is how they practice music pieces. Instead of relying only on scheduled class practice or teacher feedback, students use technology to self-record and review. For instance, some described recording themselves using smartphone voice memos or video, then critiquing their performance (or even using software that provides feedback). They also engage in iterative learning where they pause and replay tutorial videos until they master a section. This self-paced, tech-enabled practice allows them to progress at their own speed. One student shared: *"I learned a complex drum pattern by slowing down a YouTube video to 0.5x speed and playing along until I nailed it."* Which is different aspect of learning music.

3. Peer-to-Peer Collaboration (Online and Offline): Digital native learners often form communities, both virtual and face-to-face, to aid each other. In the focus groups, students gave examples such as exchanging tutorial recommendations in group chats, collaborating on cloud-based projects (like jointly editing a composition on Google Drive or a DAW project via Dropbox), and using social media to showcase their work and get feedback. During the

pandemic, some student ensembles continued to “jam” by recording parts separately and merging them – a strategy facilitated by technology.

4. Blending Formal and Informal Learning: A defining strategy among students is blending formal and informal music learning. They view classroom instruction and self-directed exploration as complementary. For instance, after a piano lesson on scales, a student might use a gamified app to reinforce practice, while another may study chord progressions in theory class after learning a pop song by ear. This two-way interaction deepens understanding and motivation. Some instructors support this approach, inviting students to perform pieces learned independently to validate informal learning. By merging formal fundamentals with informal creativity, students gain both technical credibility and personal autonomy, maximizing learning across structured and self-directed contexts.

5. Adapting Learning Techniques: Digital natives also adapt classic learning techniques to the digital realm. For instance, flashcards for music theory now take the form of apps like Theory Lessons and Tenuto from www.musictheory.net. Metronome and tuner – once physical devices – are used as phone apps. Some students utilize online forums to ask questions when they hit a wall in self-study.

6. Seeking Mentorship and Micro-Learning: Interestingly, while these students are self-directed, they still value mentorship – but they often seek it outside the local context. A few participants mentioned that they follow famous musicians or educators on social media, essentially treating them as virtual mentors. For example, composition students might follow film composers on Twitter who often give tips or host Q&A, or an aspiring DJ follows a producer’s Twitch livestream to glean production tricks. This strategy of micro-learning from experts around the world is unique to the digital generation. It expands

Overall, the action/interaction strategies of digital native learners are multifaceted and dynamic. They seize control of their learning, employ technology to access information, practice efficiently, collaborate, and personalize their education. These strategies illustrate agency – students making decisions about how, when, and with whom to learn – enabled by the affordances of the digital age.

Consequences (Outcomes): The theory would be incomplete without examining the outcomes – what results from this new mode of learning, and what are its implications? Our

findings indicate several consequential outcomes at both the student and Elaborate on the impact on students at the institutional level:

1. Enhanced Skills and Creativity: Many students demonstrated advanced skills and broadened musicality through blended learning. Instructors noted that some surpassed class expectations, such as a guitarist who mastered fingerstyle techniques from YouTube over the summer. This reflects broader trends among digital-native learners who achieve high proficiency through self-directed study. Exposure to diverse online content also fosters versatility and creativity, encouraging experimentation in composition, improvisation, and cross-genre projects. One student, for example, applied knowledge from a free online production course to create an original EDM track blending classical Thai instruments with electronic beats—showing how digital resources expand both skill and artistic innovation.

2. Increased Learner Autonomy and Confidence: Another outcome is that students develop a sense of autonomy in learning, which can last into their professional lives. By taking charge of their education, they build confidence in their ability to acquire new skills on their own. Focus group participants expressed pride in being self-starters. “I feel I can learn anything if I want to,” said one student, reflecting a growth mindset. This confidence also means they may require less handholding and can be proactive in problem-solving – a positive trait in higher education and beyond. However, instructors noted that this autonomy sometimes challenges the traditional teacher-centered model. A few instructors felt that highly autonomous students could appear “less respectful” or question the teachers more. This points to a cultural shift in teacher-student dynamics: the consequence of increased autonomy is a need for educators to adapt, moving toward a mentor or coach role rather than a top-down authority.

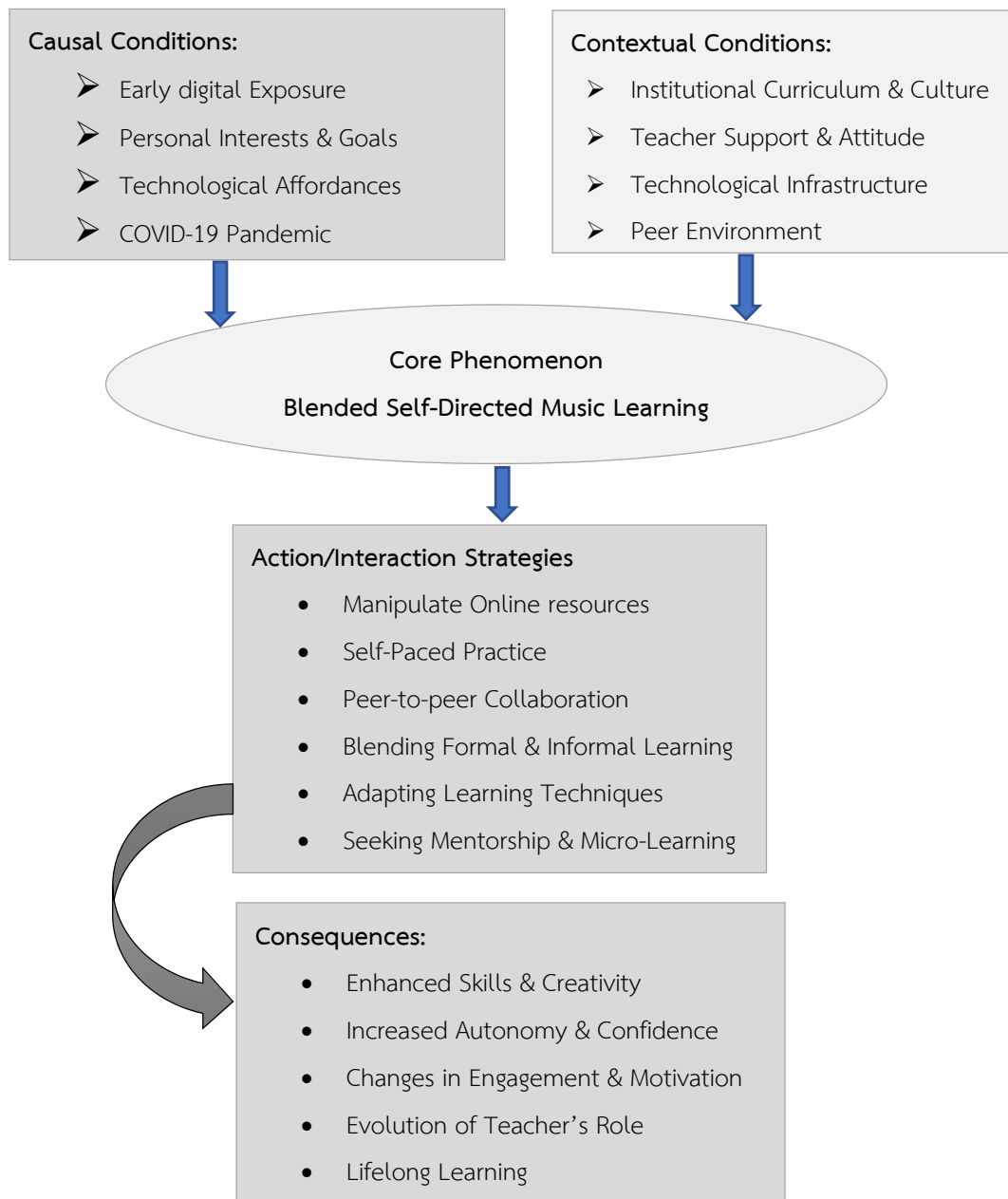
3. Changes in Engagement and Motivation: Generally, the blended learning model leads to increased engagement because students are pursuing what genuinely interests them alongside required work. Many students reported that using technology (apps, videos) made practice more enjoyable, which in turn made them more likely to practice longer or more often than if they only had traditional methods. For example, gamified elements (earning points in an app, or the fun of playing along with a backing track from YouTube) can sustain motivation. However, a few negative consequences were also mentioned: some students

admitted that the constant availability of entertainment online can distract from disciplined practice – e.g., intending to watch one tutorial but then procrastinating with unrelated videos. Additionally, the quick answers available online might lead to superficial learning at times. An instructor observed that “because answers are a click away, they sometimes don’t deeply internalize fundamentals – they can find the chord chart quickly instead of training their ear.” This indicates a need for balance; while motivation and breadth increase, there might be a trade-off in depth or patience. It’s a consequence educator should be mindful of the “instant gratification” culture could clash with the slow, repetitive nature of mastering an instrument.

4. Evolution of the Teacher’s Role: A broader consequence, hinted above, is that the role of the teacher is evolving in this paradigm. In our model, the teacher is no longer the sole fountain of knowledge but becomes a facilitator, advisor, and curator of learning experiences. Teachers who embraced this found new ways to add value – for instance, helping students critically evaluate the quality of online content (since not all YouTube tutorials are accurate), or teaching students how to practice effectively with technology. One instructor said, “I realized my job is shifting to teach them how to learn, not just what to learn.” This is a profound shift in higher education pedagogy, aligning with calls in the literature for educators to remain relevant to youth culture and technological trends. Because of digital native learning, educators and institutions are prompted to update curricula (e.g., including modern genres or music technology courses) and invest in infrastructure (better Wi-Fi, music software, smart classrooms). At least one program, in response to student interest, recently introduced an elective on “Contemporary Digital Music Making.”

5. Lifelong Learning Route: Finally, an important long-term consequence is that these students are likely to continue learning beyond formal education in a self-directed manner. By mastering the art of learning how to learn (with the help of digital tools) during their university years, they become equipped for lifelong learning in music and other domains. In the interviews, several students aspired to careers like music production, performance, or teaching, and they acknowledged that they would need to keep updating their skills. They felt confident that they could do so via online courses or self-study. This bodes well for their adaptability in a fast-changing musical landscape.

Figure 1 Grounded Theory: Blended Self-Directed Music Learning Chart



In summary, the results of this study can be visualized as an interconnected model of causes, contexts, actions, and outcomes that define music learning for digital native students in Thai higher education. Figure 1 presents this grounded theory model, highlighting how the elements fit together. Figure 1 Grounded theory conceptual model of music learning among digital native (digital citizen) learners in Thai higher education. The model illustrates how

causal conditions (e.g., early digital exposure, personal interests, technological affordances) give rise to the core phenomenon of Blended Self-Directed Music Learning. This phenomenon unfolds within a specific context (educational and technological environment), which can facilitate or constrain the process. Students and instructors employ various action/interaction strategies (leveraging online resources, peer collaboration, blending formal/informal learning) to engage with music. These strategies lead to consequences or outcomes, including enhanced skills, increased learner autonomy.

Discussion

The grounded theory developed in this study provides a comprehensive framework for understanding how digital native learners approach music education. This framework resonates with broader observations in music pedagogy and generational studies. A central feature of the findings is the proactive, self-directed behavior of students in using technology to learn, which affirms the description of “digital natives.”⁸ Participants demonstrated fluency in digital tools, multitasking capacity, and preference for immediate information access, qualities that identify as emblematic of this generation.⁹ In line with characterization of Generation Z,¹⁰ the learners in this study also displayed pragmatism and autonomy such as consulting online tutorials.

1. Technology and Informal Music Learning

The strategies identified echo themes in the literature on informal music learning and digital media,¹¹ for instance, documented the role of YouTube communities in folk music

⁸ Marc Prensky, 1.

⁹ John Palfrey and Urs Gasser, *Born Digital: Understanding the First Generation of Digital Natives* (New York: Basic Books, 2010): 15.

¹⁰ Tracy Francis and Fernanda Hoefel, “‘True Gen’: Generation Z and Its Implications for Companies,” *McKinsey & Company*, accessed May 7, 2024, <https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/true-gen-generation-z-and-its-implications-for-companies>.

¹¹ Janice Waldron, “YouTube, Fanvids, Forums, Vlogs and Blogs: Informal Music Learning in a Convergent On- and Offline Music Community,” *International Journal of Music Education* 31, 1 (2013): 102, <https://doi.org/10.1177/0255761411434861>.

learning. Our findings extend this by showing how pervasive and normalized these practices have become within higher education. Mobile applications were developed for guided listening, which improved students' engagement and strategic learning¹² as found. Similarly, participants in this study employed applications and digital tools to enhance their practice routines, such as slowing playback or looping sections—techniques that amount to new forms of musical rehearsal. Such behaviors underscore that technology is not merely a source of distraction, as some educators fear, but can function as an effective pedagogical aid. However, the efficacy of these tools is not universal; disparities in digital infrastructure—such as unstable internet access or limited quality devices—pose significant barriers to equitable implementation, potentially stalling the progress of students who lack the necessary hardware or connectivity. Addressing these gaps is essential to support the vision of a “transformative digital world” for music,¹³ where digital tools enable creativity for all learners regardless of their socioeconomic background.

2. Teacher Roles and Institutional Adaptation

Another key dimension of this research concerns the evolving role of the teacher and institution. Evidence suggests that instructors who adopt facilitative approaches and embrace technology significantly enhance student learning outcomes. This aligns with contemporary pedagogical models such as the flipped classroom, where students consume instructional content outside the classroom and devote in-class time to higher-order activities. In the context of music education, Elliott's praxial philosophy emphasizes active music making and problem-solving under teacher guidance. Digital native learners extend this praxis by incorporating online resources as part of their practice. Instructor feedback in this study indicated that when teachers curated online materials and coached students on independent learning strategies.

Resistance among some instructors also emerged. Certain educators initially viewed platforms such as YouTube as a threat, fearing loss of pedagogical control or the spread of poor practices. Yet many came to recognize that students continue to need expert guidance

¹² Sunggi Cho, Yoomee Baek, and E. J. Choe, 132.

¹³ Danny Bakan and Peter Gouzouasis, 2.

to navigate online content effectively. This realization aligns with UNESCO's frameworks that describe teachers in the digital age as "learning consultants."¹⁴ Educators thus remain indispensable, but their roles increasingly involve curation, coaching, and scaffolding, greater validity.

3. Cultural Interplay: Global and Local Dynamics

A further point concerns the cultural interplay between global digital culture and local educational traditions. Thai higher music education is rooted in both Western classical and Thai classical traditions, generally delivered through hierarchical, formal methods. By contrast, digital natives belong to a global youth culture where learning is democratized, and interest driven. This tension creates challenges, such as diminished student interest in traditional repertoire, but also opportunities for innovation. Students in this study blended Thai instruments with digital loops, generating hybrid forms that embody both local and global influences. However, the new media facilitates shared creativity and dissolves boundaries between musical styles. Rather than threatening tradition, digital engagement may support cultural preservation by recontextualizing traditional forms in modern practices.¹⁵

4. Implications for Music Education

The grounded theory developed here yields several practical implications:

4.1 Curriculum Design. Flexible, student-centered curricula that incorporate learners' interests are essential. Elective modules on popular music, technology, or songwriting could complement classical training. Without formal provision, students pursue these areas independently, often without guidance. Assignments that require analysis of pieces learned informally could help bridge informal and formal domains.

4.2 Teaching Strategies. Blended pedagogy is critical. Teachers can curate quality YouTube tutorials, share resources via learning management systems, and encourage the use of practice applications. Instruction in digital literacy for music—such as critically evaluating online content or employing digital audio workstations—can strengthen students'

¹⁴ UNESCO, "Digital Learning and Transformation of Education," accessed August 1, 2025, <https://www.unesco.org/en/articles/digital-learning-and-transformation-education>.

¹⁵ Peter Gouzouasis and Danny Bakan, 11.

independent strategies. Recognizing and validating out-of-class learning fosters positive classroom climates. Activities like “show-and-tell” sessions encourage peer learning and integrate informal experiences into formal pedagogy.

4.3 Institutional Support. Infrastructure such as reliable internet, digital devices, and subscriptions to music platforms is essential. Institutions should also provide faculty workshops on emerging technologies. Moreover, assessment systems should account for informal learning achievements, motivating students to engage more deeply with self-directed efforts.

4.4 Student Guidance. Institutions must help students balance the strengths and risks of digital learning. Time management and reflective practices should be emphasized. For example, practice logs documenting both formal and informal activities can cultivate metacognitive awareness. This helps mitigate risks of superficial learning or neglect of fundamentals.

4.5 Policy Implications. At a policy level, these insights could inform national strategies for digital education. The Thai Ministry of Education, for instance, might develop resources for integrating technology into music curricula or fund Thai-language digital content creation. Beyond music, this model of blended self-directed learning could guide pedagogy across other disciplines, given Gen Z’s preference for interactive and personalized education.

5. Limitations and Future Research

Several limitations should be acknowledged to contextualize the findings of this study. First, the small instructor sample size ($n=4$) limits the representativeness of the faculty perspective; including administrators or policymakers in future studies could provide a more systemic view of institutional readiness. Second, the findings are culturally and geographically specific. The unique interplay between Thai educational traditions and the specific digital landscape in Thailand may not generalize internationally. Furthermore, the disparity in digital infrastructure—specifically unstable internet access and varying device quality—served as a confounding variable that may have influenced student engagement levels beyond the pedagogical design itself. Third, the reliance on self-reported data may introduce participant bias. While this was mitigated through the triangulation of instructor interviews and digital

artifacts, future research could benefit from employing observational methods or objective log-based data from Learning Management Systems (LMS) to enhance validity.

Regarding future inquiry, there is a clear need for quantitative studies to test the statistical relationships between the use of online resources, intrinsic motivation, and teacher support. Longitudinal designs would also be invaluable for tracking the evolution of “digital citizen” learning habits over an extended academic career.

6. Additionally, the following areas warrant further exploration

6.1 The Evolution of Instructional Roles: Future grounded theories should examine how instructors specifically adapt to the shift from knowledge providers to facilitators, particularly within the traditional “master-apprentice” frameworks of Thai music education.

6.2 Technological Equity: Research should investigate how institutionalized digital support can bridge the gap created by infrastructure disparities to ensure equitable music learning.

6.3 Emerging Technologies: The potential of Artificial Intelligence (AI), Virtual Reality (VR), and collaborative platforms deserves rigorous study to explore their capacity for enhancing creativity and social music-making.

By addressing these gaps, future scholarship can contribute to a more holistic and equitable framework for music education in the digital age.

Conclusion

This study examined music learning among digital-native students in Thai higher education, developing a grounded theory of Blended Self-Directed Music Learning. Students combine technology and personal agency to enhance formal instruction, creating individualized learning pathways. The theory identifies causal conditions (digital upbringing, personal interests, online resources), contextual factors (curriculum, teacher roles, infrastructure), strategies (tutorials, apps, peer collaboration, blending formal and informal learning), and consequences (skill development, autonomy, and innovative pedagogy). Learning now extends beyond classrooms; digital tools are integral, not distracting. Educators should guide critical engagement and reflection while embracing technology. For Thai higher education, this framework informs

curricular reform and contributes globally to understanding how Gen Z music learners construct knowledge in a digital era.

Declaration of Generative Artificial Intelligence in the Writing Process

During the preparation of this manuscript, the authors used Gemini 2.5 (Google) solely to improve language clarity and grammar. All scientific content, interpretations, analyses, and conclusions were developed by the authors. The authors reviewed and edited the AI-generated output as necessary and take full responsibility for the accuracy and integrity of the manuscript.

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